

Agent Failure Modes and Evaluation

Evaluation is about detecting failures. The more complex a task an agent performs, the more possible failure points there are. Other than the failure modes common to all AI applications discussed in Chapters 3 and 4, agents also have unique failures caused by planning, tool execution, and efficiency. Some of the failures are easier to catch than others.

To evaluate an agent, identify its failure modes and measure how often each of these failure modes happens.

I created a simple benchmark to illustrate these different failure modes that you can see on the book's [GitHub repository](#). There are also agent benchmarks and leaderboards such as the [Berkeley Function Calling Leaderboard](#), the [AgentOps evaluation harness](#), and the [TravelPlanner benchmark](#).

Planning failures

Planning is hard and can fail in many ways. The most common mode of planning failure is tool use failure. The agent might generate a plan with one or more of these errors:

Invalid tool

For example, it generates a plan that contains `bing_search`, but `bing_search` isn't in the agent's tool inventory.

Valid tool, invalid parameters.

For example, it calls `lbs_to_kg` with two parameters. `lbs_to_kg` is in the tool inventory but requires only one parameter, `lbs`.

Valid tool, incorrect parameter values

For example, it calls `lbs_to_kg` with one parameter, `lbs`, but uses the value 100 for `lbs` when it should be 120.

Another mode of planning failure is goal failure: the agent fails to achieve the goal. This can be because the plan doesn't solve a task, or it solves the task without following the constraints. To illustrate this, imagine you ask the model to plan a two-week trip from San Francisco to Hanoi with a budget of \$5,000. The agent might plan a trip from San Francisco to Ho Chi Minh City, or plan a two-week trip from San Francisco to Hanoi that will be way over the budget.

A common constraint that is often overlooked by agent evaluation is time. In many cases, the time an agent takes matters less, because you can assign a task to an agent and only need to check in when it's done. However, in many cases, the agent becomes less useful with time. For example, if you ask an agent to prepare a grant proposal and the agent finishes it after the grant deadline, the agent isn't very helpful.

An interesting mode of planning failure is caused by errors in reflection. The agent is convinced that it's accomplished a task when it hasn't. For example, you ask the agent to assign 50 people to 30 hotel rooms. The agent might assign only 40 people and insist that the task has been accomplished.

To evaluate an agent for planning failures, one option is to create a planning dataset where each example is a tuple (task, tool inventory). For each task, use the agent to generate a K number of plans. Compute the following metrics:

1. Out of all generated plans, how many are valid?
2. For a given task, how many plans does the agent have to generate, on average, to get a valid plan?
3. Out of all tool calls, how many are valid?
4. How often are invalid tools called?
5. How often are valid tools called with invalid parameters?
6. How often are valid tools called with incorrect parameter values?

Analyze the agent's outputs for patterns. What types of tasks does the agent fail more on? Do you have a hypothesis why? What tools does the model frequently make mistakes with? Some tools might be harder for an agent to use. You can improve an agent's ability to use a challenging tool by better prompting, more examples, or fine-tuning. If all fail, you might consider swapping this tool for something easier to use.

Tool failures

Tool failures happen when the correct tool is used, but the tool output is wrong. One failure mode is when a tool just gives the wrong outputs. For example, an image captioner returns a wrong description, or an SQL query generator returns a wrong SQL query.

If the agent generates only high-level plans and a translation module is involved in translating from each planned action to executable commands, failures can happen because of translation errors.